# Bonds, Forces and Properties Slides · run-sheet

## Period 1 · 45.5 min

### 1. Bonds, forces and properties Lesson 4.5 · Reading §4.5 · 0.5 min

0. **enter** (0.5 min): Two periods. Read the driving question aloud; students copy it. Show the four clicker buttons once, so a zoom or a blank screen is on purpose. Thread of the lesson: what comes apart, what stays.

**Safety.** Nothing is demonstrated in this lesson. Dry ice and molten salt are only described: dry ice (-78.5 °C) burns skin and its gas can build up in a closed space; molten salt is above 801 °C. If a real conductivity tester is used, it is a low-voltage battery tester, and the salt water stays away from wall outlets. **Data and sources.** Melting and boiling points at 1 atm: NIST WebBook and PubChem ($CH_4$ -182.5 and -161.5 °C; $F_2$ -188.1, $Cl_2$ -34.0, $Br_2$ 58.8, $I_2$ 184.4, $NH_3$ -33.3, HCl -85.1, $H_2O$ 100.0 °C; NaCl 801 °C; Cu 1085 °C; quartz near 1700 °C; $CO_2$ sublimes at -78.5 °C). CRC Handbook: densities at 0 °C, ice 0.9167 g/cm³, water 0.99984 g/cm³; calcium carbonate is sparingly soluble. OpenStax Chemistry 2e 10.1 (intermolecular forces), 10.3 (phase transitions: 40.7 kJ to boil 18 g of water), 7.5 (bond strength: about 928 kJ to break its O–H bonds), 10.5 (the solid state). Radii: covalent, Cordero 2008 (ball-and-stick); van der Waals, Alvarez 2013 (space filling); ions, Shannon 1976 ($Na^+$ 102 pm, $Cl^-$ 181 pm; NaCl spacing 282 pm); Cu 128 pm metallic radius. Ice: O···O 276 pm, ideal hexagonal oxygen framework, hydrogen directions simplified. Fritz London named dispersion forces in 1930. Electronegativity: Pauling values (H 2.20, Br 2.96). **Clicker.** Logitech R800: Next = PageDown, Back = PageUp (while zoomed: exit the zoom), Start/stop = F5 then Escape, Blank = period. Blank zooms one part of the slide per press, animation first; the press after the last part exits; two fast presses on an unzoomed slide blank the screen. Mouse: click an option to mark it right or wrong; click the slide number or the section label for the slide list. Laptop: N notes, O overview, T theme, Ctrl+P run-sheet.

### 2. Shake the bag Hook · Reading §4.5.1 · 3 min

0. **enter** (0.5 min): Everyday scene: three small brick models in a bag. Ask: who has built with snap-together bricks? No chemistry yet.
1. **predict** (0.75 min): Prediction: take one answer per table. Collect "the bricks come off" and "the models separate". Do not confirm.
2. **shake** (0.5 min): The models tumble apart; the counter stays at 11 of 11. Ask: what came apart, and what stayed?
3. **pour** (0.5 min): Spread far apart, every model still whole. Misconception to listen for: "shaking breaks the models" (it only separates them).
4. **the two words** (0.75 min): The two ideas of the lesson in kid words: a snap is a bond within; side by side is an attraction between. Students write both lines.

Narrative thread for 4.5: what comes apart, what stays. The bricks are atoms, a snap is a bond within, a whole model is a molecule; models that sit side by side only touch, an attraction between. Slide 6 names both.

### 3. Snow day Launch · Reading §4.5 opener · 3.5 min

0. **enter** (0.25 min): A snowman: water molecules. Ask: where does a snowman go on a sunny day?
1. **salt** (0.25 min): Salt on the steps: $Na^+$ and $Cl^-$ ions drawn to scale. It is there all year.
2. **sand** (0.25 min): Sand in the sandbox: silicon and oxygen atoms in one network. It never melts in any weather.
3. **bonds** (0.25 min): Bonds hold every one of the three solids together. Yet they melt at very different temperatures.
4. **predict** (0.5 min): Prediction: what comes apart when snow melts? Collect two answers. Listen for "the water molecules break" (they do not).
5. **melt the snow** (0.25 min): The water molecules let go of each other and drift, each one whole. Salt and sand do not move.
6. **the numbers** (0.25 min): Melting points: snow 0.0 °C, salt 801 °C, sand near 1700 °C. Ask: which forces must come apart in each?
7. **driving question** (0.5 min): Read the driving question; students copy it at the top of the notebook page.
8. **write 60 s** (1 min): Sixty seconds with the starter. Collect two answers; do not correct yet. Listen for "salt and sand have stronger bonds" and for "nothing breaks when snow melts".

The reading opener: methane melts at -182.5 °C, salt stays solid until 801 °C, quartz (the main mineral in sand) melts near 1700 °C. "The difference lies in what must come apart when each solid melts." Snow is water molecules; salt is $Na^+$ and $Cl^-$ ions (red and blue); sand is a network of Si and O atoms (CPK colors), hence the caption.

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### 4. Predict now Predict · Reading §4.5 Before you read · 2 min

0. **enter** (0.5 min): Count hands for A, B, C and write the counts on the board. Do not react to the choices. A mouse click on an option only highlights it; there is no grading on this slide.
1. **question 2** (0.75 min): Count hands. Many choose A: "it is made of ions, so it conducts".
2. **question 3** (0.75 min): Count hands. Many choose A: the word "bond" in the name. Keep the counts for slide 25.

No answers on this slide. Students keep their letters; slide 25 returns all three answers.

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### 5. Targets and words Targets · Reading §4.5 Vocabulary · 2.5 min

0. **enter** (1 min): Read the five targets. Students copy 4.5.1 to 4.5.5; they rate themselves on slide 26.
1. **words** (1 min): Choral reading: English term, then the Spanish cognate. Students add the words to the glossary with the icon. "Intermolecular" means between molecules.
2. **color code** (0.5 min): Same color code as Lessons 4.1 and 4.2. New: the solid stick (bond within) and the dashed line (attraction between).

The color code is the unit code of Lessons 4.1 and 4.2, plus two marks for this lesson: a solid gray stick is a bond within a molecule, a dashed line is an attraction between molecules. Dispersion lines are gray, dipole-dipole lines run from δ+ (warm) to δ− (teal), hydrogen bonds are teal and heavier. Ball-and-stick atoms use CPK colors with the symbol on every sphere.

## 6. Bonds within, attractions between Bonds and attractions · Reading §4.5.1 · 4.5 min

0. **enter** (0.5 min): $Cl_2$ molecules, ball-and-stick, Cl green. Ask: how many atoms in each molecule? (two).
1. **predict** (0.5 min): Prediction: hands up for "inside" or "between". Most say inside; ask why.
2. **bond within** (0.5 min): The stick inside each molecule glows: a covalent bond, a bond within.
3. **attraction between** (0.5 min): Dashed gray lines between neighbors flicker: attractions between, much weaker. Thread: in liquid chlorine these let go; the bonds stay.
4. **salt** (0.25 min): Salt: $Na^+$ 102 pm and $Cl^-$ 181 pm at 282 pm. No molecules: ionic bonds run through the whole crystal.
5. **copper** (0.25 min): Copper: gray + cations in a sea of violet - electrons (Lesson 4.1). Metallic bonds through the whole metal.
6. **rule** (1 min): Students copy the rule. Misconception to listen for: "molecules are held to each other by bonds" (only by much weaker attractions).
7. **check** (0.5 min): Hands up for A, B or C. The chlorine picture with the dashed attractions is back on screen: the check's subject.
8. **answer** (0.5 min): A is right. B (covalent bonds between the molecules): a covalent bond joins the two atoms inside one molecule. C (ionic bonds): chlorine is neutral molecules, not ions.

The TEKS call bonds "intramolecular forces", even though ionic and metallic crystals contain no molecules; the reading says "bonds within a substance" for that reason. Attractions between molecules are the "intermolecular forces" of slide 5. Atom colors are CPK (Cl green); the salt ions keep the unit code ($Na^+$ red, $Cl^-$ blue), hence the caption.

## 7. Dispersion forces Dispersion forces · Reading §4.5.1 · 5 min

0. **enter** (0.25 min): Two nonpolar molecules drawn as electron clouds with moving violet dots. Ask: are electrons ever still?
1. **predict** (0.5 min): Prediction: point to where the electrons could crowd. Collect two answers.
2. **flicker** (0.25 min): For an instant the dots crowd to one end: δ- there, δ+ at the other end. Say: "a small negative charge, for an instant".
3. **the neighbor** (0.25 min): The neighbor's cloud shifts the same way. Ask: which end of the neighbor faces the δ- end? (its δ+ end).
4. **attract** (0.25 min): The dashed gray line forms, fades and forms again: dispersion forces act between all molecules.
5. **the plot** (0.25 min): Evidence: boiling point against electrons per molecule. Read the two axes with the class.
6. **$F_2$** (0.25 min): $F_2$: 18 electrons, boils at -188.1 °C. A gas at room temperature.
7. **$Cl_2$** (0.25 min): $Cl_2$: 34 electrons, -34.0 °C. Still a gas at 25 °C.
8. **$Br_2$** (0.25 min): $Br_2$: 70 electrons, 58.8 °C: a liquid at room temperature.
9. **$I_2$** (0.25 min): $I_2$: 106 electrons, 184.4 °C: a solid at room temperature.
10. **the trend** (0.25 min): The dashed line: more electrons, higher boiling point. All four are nonpolar, so only dispersion forces differ.
11. **rule** (1 min): Students copy the rule. Misconception to listen for: "the I–I bond is stronger, so iodine boils higher" (boiling does not break the bond).
12. **check** (0.5 min): Hands up for A, B or C. The plot with $Cl_2$ and $Br_2$ is on screen.
13. **answer** (0.5 min): A is right. B ($Br_2$ is polar): both atoms are the same, so $Br_2$ is nonpolar. C (boiling breaks the Br–Br bond): boiling separates whole $Br_2$ molecules; the bond stays.

Fritz London explained these forces in 1930, so they are also called London dispersion forces. Halogens at 25 °C: $F_2$ and $Cl_2$ are gases, $Br_2$ is a liquid, $I_2$ is a solid; the plot shows why. The δ signs here last only an instant, unlike the lasting δ signs of a polar bond (slide 8).

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## 8. Dipole-dipole forces Dipole-dipole forces · Reading §4.5.2 · 4.25 min

0. **enter** (0.5 min): Two $F_2$ molecules, dashed gray lines only. Ask: polar or nonpolar? (nonpolar: same atoms).
1. **HCl** (0.5 min): HCl: δ+ on H (warm), δ- on Cl (teal), from Lesson 4.4.
2. **predict** (0.5 min): Prediction: students turn their hands like two molecules. Collect "δ+ toward δ−".
3. **turn** (0.25 min): The molecules turn so the δ+ end of one faces the δ- end of the next.
4. **face** (0.25 min): The dashed warm-to-teal line: dipole-dipole forces.
5. **compare** (0.25 min): Electrons per molecule: $F_2$ 18, HCl 18. Boiling point: $F_2$ -188.1 °C, HCl -85.1 °C.
6. **rule** (1 min): Students copy the rule. Polar molecules have both: dipole-dipole forces and dispersion forces.
7. **check** (0.5 min): Hands up for A, B or C. HCl with its δ signs is on screen.
8. **answer** (0.5 min): A is right. B ($CO_2$): linear, its two polar bonds cancel, so it is nonpolar. C ($CH_4$): nonpolar, dispersion forces only.

$F_2$ and HCl both have 18 electrons, so their dispersion forces are about equal. $F_2$ boils at -188.1 °C and HCl at -85.1 °C, about 100 °C higher: the extra pull is the dipole-dipole force. Polarity comes from Lesson 4.4 (shape and bond polarity).

## 9. Hydrogen bonding Hydrogen bonding · Reading §4.5.2 · 4.25 min

0. **enter** (0.25 min): Three points on the 10-electron line, unlabeled. Same number of electrons, so about the same dispersion forces.
1. **predict** (0.5 min): Prediction: which two boil far above methane? Collect answers with a reason.
2. **reveal** (0.25 min): $CH_4$ −161.5 °C, $NH_3$ −33.3 °C, $H_2O$ 100.0 °C. Ask: what do $NH_3$ and $H_2O$ have that $CH_4$ lacks? (H on N or O, a lone pair).
3. **hydrogen bond** (0.25 min): Two water molecules, 104.5°, the lone pairs as violet lobes. The dashed teal line runs from an H of one molecule to a lone pair on the O of the next. It is longer than the O–H stick.
4. **not a bond within** (0.25 min): Not a covalent bond: about 5 to 10% as strong. Misconception to listen for: "a hydrogen bond is a bond within the molecule".
5. **ammonia** (0.25 min): Two $NH_3$ molecules: H on N, a lone pair on N, the same kind of line.
6. **methane** (0.25 min): Two $CH_4$: the crossed-out line. H on C, no lone pair: dispersion forces only.
7. **four** (0.25 min): One water molecule with four neighbors: two through its H atoms, two through its lone pairs.
8. **rule** (1 min): Students copy the rule. Misconception to listen for: "HCl forms hydrogen bonds" (H on Cl, not N, O or F).
9. **check** (0.5 min): Hands up for A, B or C. Water with its four hydrogen bonds is on screen.
10. **answer** (0.5 min): A is right. B (methane): its H atoms are on carbon, not on N, O or F. C (HCl): polar, but its H is on Cl; no hydrogen bonds.

A hydrogen bond is an especially strong dipole-dipole attraction, not a covalent bond: only about 5 to 10% as strong. The 10-electron trio ($CH_4$ −161.5 °C, $NH_3$ −33.3 °C, $H_2O$ 100.0 °C) has equal dispersion forces, so the difference is the hydrogen bonding. The reading's "all eight" view of the plot is not shown.

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## 10. Name the attraction: I do, we do, you do Strongest attraction · Reading §4.5.2 · 4.5 min

0. **enter** (0.25 min): The three questions come first; the boxes on screen are empty. Read each title aloud.
1. **I do** (0.25 min): I do, thinking aloud: $NH_3$ is trigonal pyramidal (Lesson 4.4), so it is polar.
2. **H on N, O or F?** (0.25 min): Its H atoms are on N: yes.
3. **lone pair?** (0.25 min): N has one lone pair: yes.
4. **strongest** (0.25 min): Hydrogen bonding, along with dipole-dipole and dispersion forces.
5. **we do** (0.25 min): We do: students call "polar?" before the press. $CO_2$ is linear; the pulls cancel: no.
6. **H on N, O or F?** (0.25 min): Students call it first: $CO_2$ has no H at all.
7. **strongest** (0.25 min): A nonpolar molecule has only dispersion forces. Misconception: "$CO_2$ has polar bonds, so dipole-dipole forces" (the pulls cancel).
8. **you do** (0.25 min): You do: the chips give Br 2.96 and H 2.20. Students find the difference and the strongest attraction alone.
9. **timer 90 s** (1.5 min): Ninety seconds, silent, in notebooks. Circulate; look for "hydrogen bonding" (H on Br is not H on N, O or F).
10. **polar?** (0.25 min): 2.96 − 2.20 = 0.76: a polar bond, and HBr is polar.
11. **H on N, O or F?** (0.25 min): H is on Br, not on N, O or F: no hydrogen bonding.
12. **strongest** (0.25 min): Dipole-dipole forces. Students fix their notebook answer.

Reading §4.5.2, triplet 1. Watch $NH_3$ (polar, trigonal pyramidal, Lesson 4.4; H on N; a lone pair: hydrogen bonding, with dipole-dipole and dispersion forces). Finish it $CO_2$ (linear, the pulls cancel; no H: dispersion forces only). On your own HBr (2.96 − 2.20 = 0.76, polar; H on Br, not N, O or F: dipole-dipole forces). Honors (notes only, never assessed): for molecules of similar size the ranking is hydrogen bonding, then dipole-dipole, then dispersion ($CH_4$ −161.5, HCl −85.1, $H_2O$ 100.0 °C). It fails across sizes: $Br_2$, 70 electrons and dispersion forces only, boils at 58.8 °C, far above polar HCl. Slide 13 builds the ladder of attractions.

## 11. Molecules stay whole Melting and boiling · Reading §4.5.3 · 4.25 min

0. **enter** (0.25 min): Solid methane at its melting point. Count the C–H bonds with the class: 24 of 24 intact.
1. **predict** (0.5 min): Prediction: what comes apart when methane melts? Collect answers. Listen for "the C–H bonds break".
2. **melting** (0.25 min): The top row leaves its places first. The bond readout stays 24 of 24; only "Attractions drawn" falls.
3. **liquid** (0.25 min): Liquid: the molecules move past each other; attractions still form and break.
4. **boiling** (0.25 min): Boiling at -161.5 °C: the molecules spread far apart.
5. **gas** (0.25 min): The gas: attractions drawn 0, covalent bonds still 24 of 24. Misconception: "methane melts low because its C–H bonds are weak" (melting leaves them intact).
6. **water** (0.25 min): Water at 100.0 °C: steam is still $H_2O$ molecules, each with its two O–H bonds, 12 of 12.
7. **energy** (0.25 min): Two bars to one scale: 40.7 kJ to boil, about 928 kJ to break the bonds. More than 20 times as much.
8. **rule** (1 min): Students copy the rule.
9. **check** (0.5 min): Hands up for A, B or C. Steam is back on screen: water molecules far apart, 12 of 12 bonds intact.
10. **answer** (0.5 min): A is right. B (O–H bonds break): boiling does not break covalent bonds; steam is still $H_2O$. C (hydrogen and oxygen): boiling is a change of state, not a reaction.

The reading's heating model: six molecules from solid to gas; the readout "Covalent bonds: 24 of 24 intact" (water: 12 of 12) never changes, while "Attractions drawn" falls to 0 in the gas. Energy (OpenStax): boiling 18 g of water at 100 °C takes 40.7 kJ; breaking every O–H bond in it would take about 928 kJ, more than 20 times as much. Shown as the reading's comparison only; no bond-energy calculation.

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## 12. Why ice floats Ice · Reading §4.5.3 · 4 min

0. **enter** (0.25 min): The lattice turns slowly. Point at one molecule: two O–H sticks, still there in ice.
1. **hydrogen bonds** (0.25 min): The dashed teal lines: hydrogen bonds, each from an H to a lone pair on the next O.
2. **four neighbors** (0.25 min): One oxygen ringed with its four neighbors, 276 pm away.
3. **predict** (0.5 min): Prediction: more space or less when ice melts? Most say more; collect the reasons.
4. **open rings** (0.25 min): The view tips down the axis: open six-sided rings, like tunnels. Empty space inside the solid.
5. **ice floats** (0.25 min): The block floats with 8.3% above the surface. Bars: 100.0 $cm^3$ of water becomes 109.1 $cm^3$ of ice.
6. **pipes** (0.25 min): Water expands as it freezes, so it can split rocks and burst pipes. The case file returns to this.
7. **rule** (1 min): Students copy the rule. Misconception to listen for: "ice is heavier, so it should sink".
8. **check** (0.5 min): Hands up for A, B or C. The volume bars are on screen.
9. **answer** (0.5 min): A is right. B (100.0 $cm^3$): the mass stays the same, but ice is less dense, so the volume grows. C (about 92 $cm^3$): most solids shrink when they freeze; ice expands.

Hexagonal ice (data notes: ideal oxygen framework, O···O 276 pm, hydrogen directions simplified). Densities at 0 °C (CRC): ice 0.9167 $g/cm^3$, liquid water 0.99984 $g/cm^3$; a floating block shows 8.3% above the surface; 100.0 $cm^3$ of water becomes 109.1 $cm^3$ of ice, the same 99.98 g. Melting collapses the open rings, so liquid water is denser. The Texas freeze case file uses this slide's model for the burst pipes of February 2021.

### 13. One picture: the ladder of attractions Summary · Reading §4.5.1 to §4.5.3 · 3.25 min

0. **enter** (0.5 min): Bottom rung: dispersion forces, between all molecules; more electrons, stronger. Methane boils at −161.5 °C.
1. **dipole-dipole forces** (0.5 min): Middle rung: dipole–dipole forces, between polar molecules. HCl boils at −85.1 °C.
2. **hydrogen bonding** (0.5 min): Top rung: hydrogen bonding, H on N, O or F pulled to a lone pair. Water boils at 100.0 °C.
3. **the thread** (0.25 min): The thread of period 1: what comes apart is the attractions; what stays is the bonds.
4. **write 90 s** (1.5 min): Ninety seconds of writing with the frame and the word bank. End of period 1: collect three notebooks as a quick check.

One picture closes period 1: the three attractions as rungs, built from the bottom. Students copy the ladder. Honors (notes only): the order holds for molecules of similar size; across sizes it fails ($Br_2$ 58.8 °C, dispersion forces only, boils far above polar HCl at −85.1 °C). Period 1 ends here.

## Period 2 · 47 min

### 14. Warm-up Warm-up, period 2 · Reading §4.5.1 to §4.5.3 · 2.5 min

0. **enter** (0.25 min): Students read the three questions while you take attendance.
1. **timer 90 s** (1.5 min): Ninety seconds, in notebooks.
2. **answer 1** (0.25 min): Dispersion forces act between all molecules, polar or not.
3. **answer 2** (0.25 min): N, O or F, with a lone pair on N, O or F in the next molecule.
4. **answer 3** (0.25 min): Each $H_2O$ molecule stays whole with its two O–H bonds; only the attractions let go.

Period 2 starts here. The three questions retrieve 4.5.1, 4.5.2 and 4.5.3.

## 15. Does it conduct? Conductivity · Reading §4.5.4 · 5 min

0. **enter** (0.25 min): The tester: battery, switch, bulb, beaker with two electrodes. Thread for period 2: what can move?
1. **solid salt** (0.5 min): Prediction: hands up for "lights" or "stays dark". Many say lights: "salt is made of ions".
2. **close the switch** (0.25 min): The switch closes. The bulb stays dark.
3. **look inside** (0.25 min): Magnified: $Na^+$ and $Cl^-$ locked in the crystal. Charged, but they cannot move.
4. **molten salt** (0.25 min): Molten salt at 801 °C. Hands up again: lights or dark?
5. **close the switch** (0.25 min): The bulb lights.
6. **look inside** (0.25 min): The same ions, now free to move. Moving ions carry the current.
7. **salt water** (0.25 min): Salt water lights: $Na^+$ and $Cl^-$ move among water molecules. Salt is an electrolyte.
8. **copper** (0.25 min): Copper lights: delocalized electrons stream one way; the cations stay (Lesson 4.1).
9. **sugar water** (0.25 min): Sugar water stays dark: whole sugar molecules, no charge. Misconception: "anything dissolved conducts".
10. **distilled water** (0.25 min): Distilled water stays dark: very few ions. Tap water has more ions and conducts a little.
11. **rule** (1 min): Students copy the three lines. Misconception to listen for: "charge alone makes it conduct" (the charges must move).
12. **check** (0.5 min): Hands up for A, B or C. Molten salt with its moving ions is back on screen.
13. **answer** (0.5 min): A is right. B (melting makes ions): solid NaCl is already $Na^+$ and $Cl^-$, locked in place. C (melting frees electrons): NaCl has no electron sea; moving ions carry its current.

The reading's conductivity table: solid NaCl ($Na^+$ and $Cl^-$ locked: dark), molten NaCl (lights), salt water (lights), sugar water (no charged particles: dark), copper (delocalized electrons: lights), distilled water (very few ions: dark). A substance that forms ions in water, like NaCl, is an electrolyte. The magnified circle draws the ions to scale (Shannon); water molecules in CPK colors share the circle with the ions, hence the caption. Nothing is demonstrated: molten salt is above 801 °C.

## 16. Predict the bulb: I do, we do, you do Conductivity · Reading §4.5.4 · 5 min

0. **enter** (0.25 min): The four steps come first: type, particles, free to move, prediction. I do: molten KCl.
1. **I do: step 1** (0.25 min): Type: K is a metal, Cl a nonmetal: ionic.
2. **I do: step 2** (0.25 min): Particles: $K^+$ and $Cl^-$.
3. **I do: step 3** (0.25 min): Free to move? Yes: it is a liquid.
4. **I do: step 4** (0.25 min): Prediction: conducts.
5. **close the switch** (0.25 min): The bulb lights, as predicted.
6. **we do: step 1** (0.25 min): We do: solid $CaCl_2$. Students call each step before the press. Type: ionic.
7. **we do: step 2** (0.25 min): Particles: $Ca^{2+}$ and $Cl^-$.
8. **we do: step 3** (0.25 min): Free to move? No: locked in the crystal.
9. **we do: step 4** (0.25 min): Prediction: does not conduct. Melted or dissolved, it would.
10. **close the switch** (0.25 min): The bulb stays dark. Misconception to listen for: "it is ionic, so it conducts".
11. **you do** (0.25 min): You do: four samples. Students write how many light the bulb and what carries the current.
12. **timer 90 s** (1.5 min): Ninety seconds, alone. Circulate; look for "3 samples" (solid KCl counted) and for "electrons carry it".
13. **answer 1** (0.25 min): Two samples: KCl in water and molten KCl. Solid KCl and sugar water stay dark.
14. **answer 2** (0.25 min): The magnified view: $K^+$ and $Cl^-$ free to move carry the current.

Reading §4.5.4, triplet 2. Watch molten KCl (ionic; $K^+$ and $Cl^-$; free to move in the liquid; conducts). Finish it solid $CaCl_2$ (ionic; $Ca^{2+}$ and $Cl^-$; locked in the crystal; does not conduct; melted or dissolved, it would). On your own: solid KCl, dissolved KCl, molten KCl, sugar water: two light the bulb, carried by $K^+$ and $Cl^-$ ions that can move.

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## 17. Does it dissolve? Dissolving · Reading §4.5.4 · 4 min

0. **enter** (0.25 min): Six samples from the kitchen, the beach and the hardware store. Ask: which have you seen disappear in water?
1. **predict** (0.25 min): Prediction: students list the ones they think dissolve. Collect counts for each.
2. **the table** (0.25 min): Sodium chloride: ionic, dissolves. The ions spread out: an electrolyte.
3. **sucrose (table sugar)** (0.25 min): Sucrose: molecular, dissolves. Whole molecules spread out; no ions.
4. **calcium carbonate** (0.25 min): Calcium carbonate: ionic, yet it barely dissolves. Limestone and seashells last for years in water.
5. **iodine** (0.25 min): Iodine: molecular, dissolves slightly. The water turns pale yellow-brown.
6. **quartz** (0.25 min): Quartz: a covalent network, does not dissolve. Sand stays sand.
7. **copper** (0.25 min): Copper: metallic, does not dissolve.
8. **rule** (1 min): Students copy the rule. Misconception to listen for: "every ionic compound dissolves" ($CaCO_3$ barely does).
9. **check** (0.5 min): Hands up for A, B or C. Calcium carbonate is back in the beaker.
10. **answer** (0.5 min): A is right. B (every ionic compound dissolves): calcium carbonate is ionic, yet it barely dissolves. C (no molecular substance dissolves): sugar is molecular and dissolves easily.

The reading describes which substances dissolve and does not explain why; neither does this slide. Calcium carbonate is sparingly soluble (data notes): limestone, marble and seashells. Iodine dissolves slightly and tints the water pale yellow-brown. About 360 g of NaCl dissolves in 1 L of water (case file, only if asked). The beaker shows ions (unit code) and $I_2$ molecules (CPK), hence the caption.

## 18. Malleable or brittle Hardness · Reading §4.5.5 · 4.25 min

0. **enter** (0.5 min): Copper and salt, drawn to one scale. Both are hard: you cannot scratch either with a fingernail.
1. **predict** (0.25 min): Prediction: which one breaks under a sharp blow? Hands up for copper or salt; ask for a reason.
2. **the blow** (0.25 min): The hammer strikes the top layers of both blocks.
3. **the layers shift** (0.5 min): The top layers move one spacing. Copper: the electrons flow along. Salt: each shifted ion now sits beside an ion of the same charge; three pairs flash red.
4. **result** (0.5 min): Copper bends and stays one piece: malleable. Salt's layers push apart along a flat face: brittle, with a red crack.
5. **molecular solids** (0.25 min): Molecular solids such as iodine and dry ice are usually soft: only weak attractions hold their molecules.
6. **rule** (1 min): Students copy the rule. Misconception to listen for: "salt splits because ionic bonds are weak" (salt is hard; like charges repel).
7. **check** (0.5 min): Hands up for A, B or C. The split salt block is on screen.
8. **answer** (0.5 min): A is right. B (ionic bonds are weak): ionic bonds are strong, so salt is hard; like charges repel when layers shift. C (electrons flow out): salt has no electron sea; its ions stay charged.

The reading's hammer model (§4.5.5): both blocks to one scale, $Na^+$ 102 pm, $Cl^-$ 181 pm at 282 pm, Cu 128 pm. A metal bends because its electrons move with the shifting cations; an ionic crystal splits because shifted layers put like charges side by side. Molecular solids (iodine, dry ice) are usually soft: only weak attractions hold them. Nothing is demonstrated.

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## 19. Covalent network solids Network solids · Reading §4.5.5 · 3.5 min

0. **enter** (0.25 min): Dry ice: six $CO_2$ molecules, linear, with dashed dispersion lines between them. The thermometer reads −78.5 °C.
1. **predict** (0.25 min): Prediction: $CO_2$ and $SiO_2$ look alike on paper. Do they behave the same? Hands up yes or no.
2. **sublimes** (0.25 min): The molecules fly off whole; the dashed lines vanish; no liquid forms. That is sublimation.
3. **quartz** (0.25 min): Quartz: every Si bonded to four O, each O bridging two Si. No molecule outline anywhere.
4. **heat** (0.25 min): The thermometer passes 1000 °C and the network stays. Misconception to listen for: "all covalent solids melt low" (water melts at 0 °C, quartz near 1700 °C).
5. **diamond** (0.25 min): Diamond: a network of carbon atoms, the hardest natural substance.
6. **rule** (1 min): Students copy the rule.
7. **check** (0.5 min): Hands up for A, B or C. The quartz network is back on screen.
8. **answer** (0.5 min): A is right. B (dry ice): separate $CO_2$ molecules held by dispersion forces. C (iodine): separate $I_2$ molecules held by dispersion forces.

Dry ice sublimes at −78.5 °C: no liquid at 1 atm (never give $CO_2$ a boiling point). Quartz, $SiO_2$, melts near 1700 °C (sources 1710 to 1722 °C); its fragment is idealized with straight Si–O–Si. Most covalent network solids are very hard; diamond is the hardest natural substance. Teacher only, the exception: graphite conducts along its layers (one delocalized electron per carbon), and its layers are held by dispersion forces only, so it is soft. Atom colors are CPK: C dark gray, O red, Si tan.

## 20. Four kinds of solids Summary · Reading §4.5 Review · 2.5 min

0. **enter** (0.5 min): Molecular solids, iodine: melting overcomes only the attractions between molecules, so the melting point is low. No charged particles move: no current. Soft.
1. **ionic solids** (0.5 min): Ionic solids, salt: ionic bonds, a high melting point. Conducts when molten or dissolved. Hard, but brittle.
2. **metallic solids** (0.5 min): Metallic solids, copper: metallic bonds, a melting point that is usually high. Conducts as a solid and a liquid. Malleable.
3. **covalent network solids** (0.5 min): Covalent network solids, quartz: covalent bonds, a very high melting point. No current, very hard (graphite is the exception).
4. **snow, salt, sand** (0.5 min): The snow day returns: snow melts at 0 °C because only attractions let go; salt needs 801 °C for ionic bonds; sand near 1700 °C for covalent bonds.

The reading's Review table in one picture. The thread closes: what comes apart decides the melting point. Graphite is the exception in the last row (it conducts along its layers and is soft); teacher only.

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## 21. Checkpoint routine Checkpoint · Objectives 4.5.2, 4.5.4 and 4.5.5 · 1 min

0. **enter** (0.5 min): Explain that the routine repeats for each question.
1. **think** (0.1 min): Think: silent, no pencils.
2. **talk** (0.1 min): Talk: partner A first, then partner B.
3. **write** (0.1 min): Write: the sentence frames and word bank stay on screen.
4. **self-check** (0.1 min): Self-check: tick each criterion you met.
5. **compare** (0.1 min): Compare: read the sample answer and fix one line.

Routine for slides 22 to 24: think 30 s, talk 60 s, write 90 s, self-check, compare.

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## 22. Checkpoint 1 Checkpoint 1 · Objective 4.5.2 · 3.75 min

0. **enter** (0.25 min): Read the prompt aloud. Same electrons, so the same dispersion forces: the difference must be another attraction.
1. **think 30 s** (0.5 min): Silent thinking. No pencils yet.
2. **talk 60 s** (1 min): Partner A speaks first, then B. Listen for: "water's bonds are stronger" (the covalent bonds do not break on boiling).
3. **write 90 s** (1.5 min): Students write in notebooks with the frames; the word bank stays up.
4. **self-check** (0.25 min): Students tick each criterion they met and fix one line.
5. **sample answer** (0.25 min): Compare with the sample. Reading §4.5.2: hydrogen bonding needs an H on N, O or F and a lone pair on N, O or F.

Routine: think 30 s, talk 60 s, write 90 s, self-check, compare. The question stays on screen on every frame.

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## 23. Checkpoint 2 Checkpoint 2 · Objective 4.5.4 · 3.75 min

0. **enter** (0.25 min): Read the prompt aloud. The same ions are in both; what changes?
1. **think 30 s** (0.5 min): Silent thinking. No pencils yet.
2. **talk 60 s** (1 min): Partner A speaks first, then B. Listen for: "salt water conducts through electrons".
3. **write 90 s** (1.5 min): Students write in notebooks with the frames; the word bank stays up.
4. **self-check** (0.25 min): Students tick each criterion they met and fix one line.
5. **sample answer** (0.25 min): Compare with the sample. Reading §4.5.4: a substance conducts when it has charged particles that can move.

Routine: think 30 s, talk 60 s, write 90 s, self-check, compare. The question stays on screen on every frame.

## 24. Checkpoint 3 Checkpoint 3 · Objective 4.5.5 · 3.75 min

0. **enter** (0.25 min): Read the prompt aloud. Picture the hammer model of slide 18.
1. **think 30 s** (0.5 min): Silent thinking. No pencils yet.
2. **talk 60 s** (1 min): Partner A speaks first, then B. Listen for: "ionic bonds are weak" (salt is hard; like charges repel).
3. **write 90 s** (1.5 min): Students write in notebooks with the frames; the word bank stays up.
4. **self-check** (0.25 min): Students tick each criterion they met and fix one line.
5. **sample answer** (0.25 min): Compare with the sample. Reading §4.5.5: a metal bends because its electrons move with the shifting cations; an ionic crystal splits because shifted layers put like charges side by side.

Routine: think 30 s, talk 60 s, write 90 s, self-check, compare. The question stays on screen on every frame.

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## 25. Predictions and exit ticket Exit ticket · Objectives 4.5.1, 4.5.3 and 4.5.5 · 6 min

0. **enter** (0.5 min): Answer 1: $H_2O$ molecules. Hands up if you predicted A. Compare with the counts from slide 4.
1. **answer 2** (0.5 min): Answer 2: does not conduct; the ions are locked. Ask a student who changed their mind to say why.
2. **answer 3** (0.5 min): Answer 3: an attraction between molecules, about 5 to 10% of a covalent bond.
3. **exit ticket** (0.5 min): Exit ticket. Claim first: what decides the melting point?
4. **evidence** (0.5 min): Evidence: the snow day, with the three melting points of slide 3.
5. **write 3 min** (3 min): Reasoning: attractions are much weaker than bonds. Three minutes of writing.
6. **self-check** (0.25 min): Self-check with the three criteria.
7. **sample answer** (0.25 min): Compare with the sample. Collect the tickets.

Predictions from slide 4 come back first; students check their letters. Then the CER exit ticket on the driving question, with the snow day as evidence. The question stays on screen while the sample shows. Collect the exit tickets.

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## 26. Rate your learning Reflect · All objectives · 1.5 min

0. **enter** (0.5 min): Explain the scale: 1 not yet, 2 almost, 3 I can teach it.
1. **target 1** (0.2 min): Fingers for target 4.5.1. Note who shows 1 and point to the review slides.
2. **target 2** (0.2 min): Fingers for target 4.5.2. Note who shows 1 and point to the review slides.
3. **target 3** (0.2 min): Fingers for target 4.5.3. Note who shows 1 and point to the review slides.
4. **target 4** (0.2 min): Fingers for target 4.5.4. Note who shows 1 and point to the review slides.
5. **target 5** (0.2 min): Fingers for target 4.5.5. Note who shows 1 and point to the review slides.

Fingers 1 to 3 for each target. Students below 3 revisit the review slides named on the right.

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## 27. Next Next · Missions 4.5 and the case file · 0.5 min

0. **enter** (0.5 min): Point to the three missions. Preview the case file: water expands when it freezes, so pipes burst.

Missions for Lesson 4.5: Name the attraction (six substances), Explain the property (six properties of real materials), Rank the boiling points (five sets, lowest to highest). Next in Unit 4: the Texas freeze case file, why water pipes burst in the February 2021 freeze; it uses the ice model of slide 12.